DAY 05

Git and GitHub

Easy-to-understand notes • Commands • Practical examples

Learning goal: Understand version control, create a repository, track changes, save commits, and upload a project to GitHub.

Topic	What you will learn
Git	Track changes in project files on your computer.
GitHub	Store and share Git repositories online.
Basic commands	init, status, add, commit, push, pull, and clone.
Branches	Work on a separate line of development.
README.md	Explain what a project contains and how to use it.

Prepared for Python Internship Tasks

1. What is Git?

Git is a **version control system**. It records changes made to files so that developers can see what changed, save milestones, and work safely on projects. Git works on your computer.

Example: If you update a Python program today and it stops working, Git can help you inspect earlier versions and return to a previous version when needed.

2. What is GitHub?

GitHub is an online platform for hosting Git repositories. It helps developers back up code, share projects, collaborate with others, and show their work to recruiters or teammates.

> **Remember:** Git is the version-control tool; GitHub is an online service that can host Git repositories.

3. Git vs GitHub

Git	GitHub
Installed and used on your computer.	A website/service used online.
Tracks file changes and commits.	Hosts repositories and supports collaboration.
Can work without an internet connection.	Internet is generally needed to access the hosted repository.
Commands include git add and git commit.	Common remote actions include pushing and pulling.

4. Important Git terms

Term	Simple meaning
Repository (repo)	A project folder tracked by Git.
Working directory	Files you are currently editing.
Staging area	A place where you select changes for the next commit.
Commit	A saved snapshot of staged changes.
Branch	A separate line of work, often used for a feature or fix.
Remote	A version of the repository stored somewhere else, such as GitHub.
Push	Send local commits to a remote repository.
Pull	Fetch and integrate changes from a remote repository.
Clone	Copy an existing remote repository to your computer.

5. Install and configure Git

After installing Git, check that it is available in PowerShell or Terminal:

```
git --version
```

Set the name and email that you want Git to record on your commits:

```
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
git config --global --list
```

> Use the name and email you want associated with your commits. Do not put your GitHub password or access token in Git configuration.

6. Create a local repository and make a commit

Open PowerShell in your project folder. The following commands show the common workflow:

```
# 1. Initialize Git in the current folder
git init

# 2. Check file status
git status

# 3. Stage all changes in the current folder
git add .

# 4. Save a snapshot with a message
git commit -m "Initial project commit"

# 5. View recent commits
git log --oneline
```

> **Workflow:** Edit files → git status → git add → git commit. A commit is stored locally; it is not uploaded to GitHub until you push it.

7. Create a repository on GitHub and connect it

1. Sign in to GitHub and choose **New repository**.
2. Enter a repository name and choose Public or Private.
3. If you already have a local project with commits, avoid initializing the remote repository with an extra README to keep the first push simple.
4. Copy the repository's HTTPS URL and connect it to your local project.

```
git branch -M main
git remote add origin https://github.com/USERNAME/REPOSITORY.git
git remote -v
git push -u origin main
```

Replace USERNAME and REPOSITORY with your actual GitHub username and repository name. If a remote named origin already exists, check it with `git remote -v` instead of adding it again.

8. Common Git commands

Command	Purpose
`git status`	Shows changed, staged, and untracked files.
`git add filename.py`	Stages one file.
`git add .`	Stages changes under the current directory.

`git commit -m "message"`	Creates a commit from staged changes.
`git push`	Uploads local commits to the configured remote.
`git pull`	Downloads and integrates remote changes.
`git clone REPOSITORY_URL`	Copies a remote repository to the computer.
`git branch`	Lists local branches.
`git switch -c feature-name`	Creates and switches to a new branch.
`git switch main`	Switches to the main branch.
`git remote -v`	Displays configured remote URLs.

9. Branching basics

A branch lets you develop a feature or fix separately from the main branch. This helps keep work organized.

```
# Create a branch and move to it
git switch -c add-login-page

# Check the current branch
git branch

# Stage and commit your changes
git add .
git commit -m "Add login page"

# Switch back to main
git switch main
```

Teams often merge a completed branch into main using a pull request on GitHub. For beginner practice, understand how to create, switch, and list branches first.

10. What is README.md?

README.md is a project information file written in Markdown. GitHub normally displays it on the repository's main page. A useful README explains the project's purpose, contents, requirements, setup steps, and how to run it.

```
# Python Internship Tasks

## About
This repository contains my day-wise internship tasks and learning notes.

## Contents
- Day-01: Full Stack Development Fundamentals
- Day-02: Python Fundamentals
- Day-03: Python Object-Oriented Programming
- Day-04: Advanced Python Concepts
- Day-05: Git and GitHub

## Tools
- Python
- Git
- GitHub
- Visual Studio Code

## Author
Your Name
```

11. Practical task: upload internship work to GitHub

Use this example when your local project folder already contains the files you want to upload:

```
# Move into your project folder first
cd "C:\path\to\Python-Internship-Tasks"

# Check files and Git status
tree /F
git status

# Stage and commit files
git add .
git commit -m "Add internship task files"

# Push to GitHub (after origin is configured)
git push
```

If Git says there is nothing to commit, it usually means there are no new staged changes. Run `git status` to check before committing.

12. Troubleshooting

Message / issue	What to do
'git' is not recognized	Install Git for Windows, then close and reopen PowerShell.
nothing to commit	Run git status. Check whether files were changed or already committed.
remote origin already exists	Run git remote -v. Use git remote set-url origin URL if the URL needs correction.
authentication failed	Complete GitHub's supported browser/sign-in flow. Do not paste passwords into commands.
repository not found	Check the repository URL, account, and access permissions.
rejected push	Run git status and read the full error. If the remote has separate commits, pull and reconcile them befor

13. Quick revision

• Git tracks changes in files; GitHub hosts repositories online.

• git init creates a local Git repository.

• git add stages changes for a commit.

• git commit saves staged changes locally.

• git push uploads commits to a remote repository.

• git pull brings remote changes into the current branch.

• git clone copies an existing repository to your computer.

• A branch lets you work separately from the main branch.

• README.md describes the project and helps others understand it.

14. Practice questions

1. What is Git, and why is it used?

2. What is the difference between Git and GitHub?

3. What is a repository?

4. Explain the purpose of git add and git commit.

5. What is the difference between git push and git pull?

6. What does git clone do?

7. Why are branches useful?

8. What information should a README.md contain?

9. Write the commands to initialize a repository, add files, commit, and push to GitHub.

10. What should you check if git remote -v shows no output?

End of Day 05. Practice the commands in a test folder first. Check `git status` often, write meaningful commit messages, and never upload passwords, API keys, or private information.